

Vinh Ly

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Summary

Software Engineer with **18 years** designing and shipping distributed systems at scale. Built real-time marketplace infrastructure at Whatnot handling **\$300M+** in annual GMV with **Ruby, Go, and Kubernetes**. Owned YouTube search ranking infrastructure serving billions of queries daily. Led engineering teams at Google delivering search infrastructure powering Google's core index. Hands-on with **AWS, GCP, C++, and Python**. Seeking senior platform engineering role. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS

Distributed Systems: Go, Ruby, Python, C++, Java

Infrastructure: Kubernetes, AWS, GCP, Terraform

Data Stores: PostgreSQL, Spanner, Bigtable, Kafka

Platform Engineering: CI/CD, GitLab, monitoring, alerting, on-call

Work Experience

Staff Software Engineer, YouTube

02/2025 - Present | Remote

- Owned YouTube search ranking infrastructure serving 2+ billion daily queries with sub-100ms p99 latency requirements.
- Mentored 4 junior engineers on search infrastructure fundamentals and distributed system design patterns.
- Designed ranking features processing user engagement signals across **Spanner** and **Bigtable** backends.
- Reduced infrastructure costs by 15% through query optimization and cache layer improvements.

Staff Software Engineer, Whatnot

05/2024 - 02/2025 | Remote

- Built real-time marketplace platform processing **\$300M+** annual GMV with **Ruby, Go, and Kubernetes** on **AWS**.
- Migrated monolithic Rails services to event-driven **Go** microservices reducing p99 latency from 800ms to 120ms.
- Led on-call rotation covering platform availability for 2M+ monthly active buyers across live shopping categories.
- Established engineering standards for **PostgreSQL** schema design and query optimization across 12 product squads.
- Designed auction settlement system handling 50K+ concurrent bid streams with exactly-once delivery guarantees.

Staff Software Engineer/Manager, Google

03/2015 - 04/2024 | Remote

- Led team of 8 engineers building core search infrastructure serving 40+ billion documents with sub-second freshness requirements.
- Drove adoption of **Kubernetes** for ML training workloads cutting model iteration time from weeks to days.
- Collaborated with ranking teams to ship 3 major quality improvements increasing click-through rates by 2.3%.
- Reduced search infrastructure memory footprint by 25% through custom compression algorithms in **C++**.
- Architected distributed indexing pipeline processing 100M+ document updates daily across **Bigtable** clusters.

Software Engineer, Google

02/2011 - 02/2015 | Remote

- Developed ranking features for Google Search using **C++** and **Python** processing petabyte-scale data.
- Optimized inverted index structures reducing query latency by 18% for high-traffic queries.
- Participated in on-call rotation for search serving infrastructure with 99.99% uptime SLA.
- Built evaluation infrastructure measuring search quality improvements across 30+ international markets.

Software Design Engineer II, Microsoft

10/2007 - 12/2011 | Remote

- Automated regression test suite in **C#** reducing manual testing effort by 40% per release cycle.
- Authored test plans covering 200+ API surfaces for Visual Studio compatibility verification.
- Filed and tracked 500+ product defects through full software development lifecycle.
- Tested and validated **.NET** framework components for enterprise deployment scenarios.

Software Design Engineer Intern, Microsoft

06/2006 - 08/2006 | Remote

- Developed internal tooling in **C#** for test environment provisioning and management.
- Contributed to Windows Vista compatibility testing efforts across legacy application matrix.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007